



AlphaX PieceWork

Implementation & Administration Guide

دليل التنفيذ والإدارة — نظام أجور العمل بالقطعة

Product: alphas_piecework v1.0.0 for Frappe / ERPNext v15

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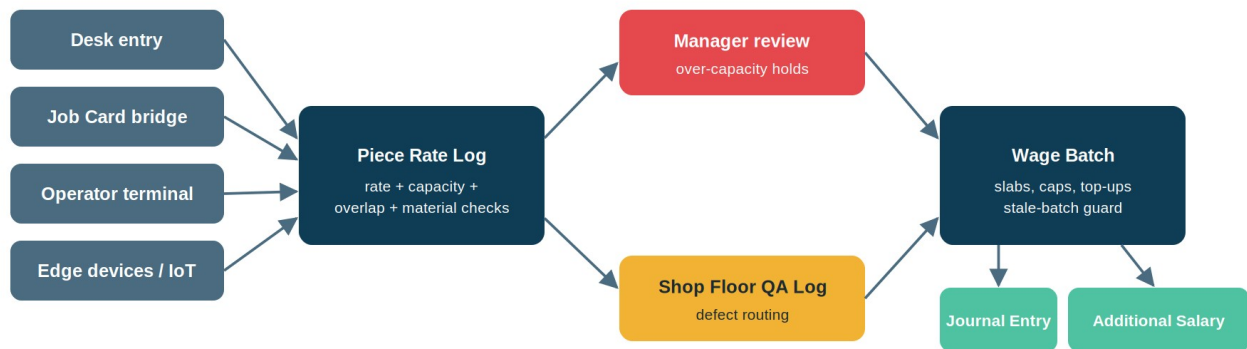
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1. What the app does

AlphaX PieceWork turns shop-floor output into correct, auditable piece-rate wages inside ERPNext. Output can be captured from the desk, from ERPNext Job Cards, from an offline-capable operator terminal, or from barcode scanners, RFID gates and PLC counters through a token-secured API. Every entry is priced from an effective-dated rate matrix, checked against plausible capacity, material issued and upstream output, adjusted for quality defects according to who was responsible, and posted to the general ledger or to HRMS payroll.



Every sensitive action appends to the hash-chained PieceWork Audit Event trail

1.1 Enhancements beyond the original development plan

Area	Original plan	What was built
Hours	Default of 8 hours when blank	Hours derived from real From/To timestamps; logs over 24 h, future times and overtime above hours are rejected
Capacity	Shift hours × machine speed, hard block	Stricter of rated speed or Standard Allowed Minutes × 130 % efficiency cap, scaled by worker skill; excess is held for manager review
Approvals	Not specified	Segregation of duties: a manager cannot approve a log they entered; supervisors cannot approve
Ghost logging	Not covered	Overlapping time windows for the same worker (including cell members) are rejected
Material check	Sum of all operations vs produced qty	Per-operation total vs Work Order qty and material transferred, under a row lock; flow balance against the previous routing step (BOM fallback)
Quality	Deduct on rejection	Responsibility routing: worker is never charged for machine, material or design faults; rework pays the specialist
Incentives	Flat slabs	Marginal (tax-bracket style) slabs on daily accepted pieces, independent of entry order
Labour law	Not covered	Configurable damage and total deduction caps plus a minimum daily earnings guarantee
Posting	JE, Additional Salary or both	One mode per batch ("both" double-books expense); HRMS optional with salary-structure pre-flight; background posting with retry
Integrity	Version history	Hash-chained audit trail with one-click verification that

Area	Original plan	What was built
		detects direct database edits
Devices	Not covered	Idempotent offline-first ingest with client UUIDs; replays after Wi-Fi drops never double count

2. Components

2.1 DocTypes

DocType	Type	Purpose
Piece Rate Matrix	Submittable	Effective-dated rates per operation, optionally narrowed by item, workstation type or difficulty; overtime multiplier, SAM, urgent and multi-skill premiums, progressive slabs
Piece Rate Log	Submittable	One unit of recorded output for a worker or production cell, with pricing, capacity result, review status and payroll status
Shop Floor QA Log	Submittable	Defect record against a log: disposition, responsibility, deductions, material recovery, rework payout
PieceWork Wage Batch	Submittable	Period payroll run: per-employee totals, line-level detail, posting result
Production Cell	Master	Team line with members, skill weights and split method
Employee Operation Skill	Master	Certified operations per employee with skill level and capacity factor
PieceWork Difficulty Class / Defect Code	Master	Bilingual rate multipliers and defect catalogue (seeded)
PieceWork Edge Device / Edge Event	Master / Log	Registered devices bound to an integration user; raw event queue
PieceWork Audit Event	Immutable log	Hash-chained record of sensitive actions
PieceWork Settings	Single	All policy switches and per-company wage accounts

2.2 Screens, reports and outputs

- **Operator Terminal** (/app/piecework-terminal): large-button tablet entry that keeps working offline and syncs every 15 seconds.
- **Shop Floor Leaderboard** (/app/shopfloor-leaderboard): full-screen EN/AR monitor view showing pieces, efficiency and quality, never money.
- **Piece Rate Earnings Register**: line-level earnings with cell logs split per member.
- **Labor Effectiveness**: availability × performance × quality by employee, workstation or operation.
- **Defect Pareto**: defects ranked with cumulative share and the vital-few flag.
- **Capacity Exceptions**: every log that exceeded its ceiling, with basis, excess % and review outcome.
- **PieceWork Wage Statement**: bilingual print format for the wage batch.

3. Installation

3.1 Frappe Cloud

1. Push the repository to GitHub (the zip root is the repository root).
2. In the bench group, add the app from the GitHub repository and deploy. HRMS is optional.
3. Install AlphaX PieceWork on the site. The installer creates roles, custom fields, seeded masters and settings defaults, and re-runs safely on every migrate.
4. Before pushing future changes, run `python3 verify_tree.py` from the repository root.

3.2 Self-hosted bench

```
bench get-app https://github.com/<org>/alphax_piecowork
bench --site <site> install-app alphax_piecowork
bench --site <site> migrate
```

3.3 What the installer adds to standard doctypes

DocType	Field	Use
Workstation	Max Pieces / Hour	Rated-speed capacity ceiling
Operation	Default SAM (minutes)	Fallback SAM when the matrix row has none
Work Order	Urgent Order	Applies the urgent premium automatically
Employee	Piece Rate Eligible, Default Production Cell	Eligibility and default team

Uninstall: removing the app drops its tables and the custom fields above. Journal Entries and Additional Salaries already posted remain in ERPNext.

4. Configuration checklist

#	Where	Setting	Recommended start
1	PieceWork Settings → Company Accounts	Wage expense, wage payable, cost center, salary component (HRMS)	Salary expense; a dedicated payable account
2	PieceWork Settings → Capture	Require time window, block overlapping logs	Both on
3	PieceWork Settings → Capacity	Basis, max efficiency %, action	Stricter of Both, 130 %, Hold for Review
4	PieceWork Settings → Material	Material balance, material transferred, flow balance, allowance %	All on; leave allowance blank to inherit Manufacturing Settings
5	PieceWork Settings → Quality	Default payout / deduction / recovery %	Scrap 0 %, third-party 100 % / 100 %
6	PieceWork Settings → Payroll	Posting mode, enqueue threshold	Journal Entry Accrual, 50 employees
7	PieceWork Settings → Compliance	Deduction caps, minimum guarantee	Caps on (5 days, 50 %); confirm with counsel

#	Where	Setting	Recommended start
8	Workstation / Operation	Max pieces per hour, SAM	From time studies
9	Piece Rate Matrix	One submitted row per operation	Add narrower rows only where rates differ
10	Roles	Manager, Supervisor, QC Inspector, Device	Device role only on integration users

5. Daily operation

5.1 Capture

Supervisors enter or confirm Piece Rate Logs. With the Job Card bridge enabled, each completed Job Card time log with an employee creates and submits a log automatically; cancelling the Job Card cancels its logs unless they are already in a wage batch. Terminal and device events accumulate into one draft log per worker, work order, operation, workstation and day; drafts from previous days are auto-submitted by the daily job so the full validation runs.

5.2 Validation on submit

1. Time window and overlap checks for the worker or every cell member.
2. Rate resolution: the most specific submitted matrix row valid on the posting date.
3. Capacity ceiling: $\text{hours} \times \text{max pieces/hour}$, or $(\text{hours} \times 60 \div \text{SAM}) \times \text{efficiency cap}$, whichever is stricter, scaled by skill factor. Cells scale by member hours.
4. Work Order checks under a row lock: operation must be in the routing; per-operation total within order qty and material transferred; not above the previous operation's accepted output.
5. Over-capacity logs are submitted as Pending Review and excluded from payroll until a different manager approves them with a reason.

5.3 Quality

Disposition	Worker responsible	Machine / material / design fault
Scrap	Loses failed-piece pay less configured payout %; optional material recovery	No deduction, no recovery
Rework - Original Worker	Loses failed-piece pay; earns configured % back for doing the rework	No deduction, no payout
Rework - Third Party	Loses configured %; specialist earns configured %	No deduction; specialist is still paid

Deductions use the realised pay per piece of the original log, so overtime and premiums are reversed proportionally. Total failed pieces can never exceed pieces logged.

5.4 Wage batch

1. Create a PieceWork Wage Batch for a closed period (optionally one department or cell) and click Fetch Entries.

2. Review the employee table: gross, slab incentive, rework earnings, QC deductions, statutory waiver, minimum top-up, net.
3. Submit. Up to the enqueue threshold, posting happens in the same transaction; larger batches post in the background. A failed background posting leaves the batch submitted with status Failed and the error text, and sources stay locked so nothing is paid twice. Fix the cause and click Retry Posting.
4. Cancelling a batch reverses the Journal Entry or Additional Salaries and releases the logs for a new batch.

Run batches on closed days. Slabs are computed on each owner's daily accepted output inside the batch. Splitting one working day across two batches under-pays the marginal slab.

6. Calculation rules with a worked example

The figures below are taken from the automated scenario that runs against a live ERPNext site. Rate for Stitching is SAR 2.00 per piece, overtime $\times 1.5$, slabs $\times 1.00$ up to 600 pieces and $\times 1.25$ above.

Step	Worker A (individual)	Amount (SAR)
Log 1	08:00–12:00, 380 pieces $\times 2.00$	760.00
Log 2	13:00–17:00 incl. 1 h overtime, 400 pieces: $300 \times 2.00 + 100 \times 2.00 \times 1.5$	900.00
Log 3	18:00–19:00, 150 pieces (ceiling 100) held, then approved by a second manager	300.00
QA 1	20 pieces failed, workmanship, third-party rework: 20×2.00 deducted	–40.00
QA 2	10 pieces failed, machine fault: worker not charged	0.00
Slab	Accepted pieces $360 + 390 + 150 = 900$; $(900 - 600) \times 2.00 \times 0.25$	150.00
Cap check	Damage 40 is below 5 days' wage and 50 % of earnings: no waiver	0.00
Net	$1,960 + 150 - 40$	2,070.00

In the same run, the rework specialist earned 40.00; a two-person cell logging 700 pieces earned 1,400 split 700/700 plus a 50.00 slab split 25/25; the batch total of 3,830.00 posted as one balanced Journal Entry.

6.1 Statutory guards

- **Damage cap:** damage and material-recovery deductions above N days' wage are waived (default 5). Daily wage is the configured fixed daily wage or the period's average daily earnings.
- **Total cap:** total deductions above X % of earnings are waived (default 50 %). The stricter cap applies.
- **Minimum guarantee:** when enabled, earnings below days worked $\times$ minimum daily earning are topped up.
- **Floor:** net pay never goes below zero; the shortfall is noted on the employee row.

7. Terminal and device integration

7.1 Set-up

1. Create a System User with only the PieceWork Device role and generate an API key and secret. Administrator cannot be bound to a device.
2. Create a PieceWork Edge Device with that user, the company, and optionally a workstation and default operation.
3. On a tablet, sign in as that user, open PieceWork Terminal and select the device once; it is remembered on the tablet.

7.2 API contract

```
POST /api/method/alphax_piecework.api.edge.ingest
Authorization: token <api_key>:<api_secret>
Content-Type: application/json
```

```
{
  "device_id": "TAB-01",
  "events": [
    {
      "client_uuid": "8f2c...",
      "employee": "HR-EMP-00012",
      "qty": 12,
      "from_time": "2026-09-10 08:00:00",
      "to_time": "2026-09-10 08:10:00",
      "work_order": null,
      "operation": null
    }
  ]
}
```

```
200 {
  "message": {
    "accepted": ["8f2c..."],
    "duplicates": [],
    "rejected": [],
    "server_time": "2026-09-11 08:00:01"
  }
}
```

- Generate a UUID per event on the device and keep it in a local queue.
- Remove an event from the queue only when its UUID comes back as accepted, duplicate or rejected. Resending is always safe.
- Up to 500 events per request by default; timestamps more than 15 minutes ahead of server time are rejected.
- Use handshake(device_id) at start-up for server time and limits, and work_orders(device_id) for open orders and their routing.
- Events are aggregated by a background job right after ingest, with an hourly fallback; processed events are purged after 30 days.

8. Security, privacy and audit

Control	Implementation
Role separation	Supervisors capture; QC inspectors record defects; managers approve holds and run payroll; device users can only call edge endpoints for their own device
Segregation of duties	Approver of a capacity hold must differ from its creator
Tamper evidence	Each audit event hashes the previous one; Verify Hash Chain reports the first broken sequence after any direct database edit
Double-payment prevention	Sources are locked to a batch at submit; a stale batch fetched before another batch was submitted is refused
Concurrency	Work Order row lock during material checks; SKIP LOCKED queue processing; audit sequence lock
Privacy (PDPL)	Leaderboard shows no monetary values and can mask employee names; wage data restricted by role

Control	Implementation
API hardening	POST-only ingest, token auth, device-to-user binding, batch size and clock-skew limits

9. Test evidence

The app was installed on a real bench (Frappe v15, ERPNext v15, MariaDB 10.11) with ERPNext set up for a Saudi company in SAR, then exercised with and without HRMS v15.

Test	Result
Pure calculation engine unit tests	18 / 18 passed
Structural guard (verify_tree.py): schemas, links, controllers, hooks, pages, reports	Passed
Install, uninstall (no leftover tables or custom fields), clean reinstall	Passed
Live end-to-end scenario on the reinstalled app, HRMS present	83 / 83 checks passed
HTTP edge API with token auth: accept, replay dedup, GET refused, no-auth refused, device user denied wage data	Passed
Hourly and daily scheduler jobs: events aggregated and auto-submitted	Passed
JavaScript syntax check of all client scripts and pages	Passed

The live scenario covers rate resolution, overtime and urgent premium, capacity holds and review permissions, overlap, QC routing, over-rejection, cells, Work Order flow and material balance (with and without a routing table), slab and net calculations, Journal Entry posting and reversal, stale-batch refusal, background failure and retry, Additional Salary with salary-structure pre-flight, the Job Card bridge, audit-chain tamper detection, edge ingest and aggregation, all four reports, the leaderboard API, the print format and both desk pages. Run it on any staging site with:

```
bench --site <staging-site> execute alphax_piecowork.tests.e2e_scenario.run
```

Staging only. The scenario writes and commits its own tagged data and refuses to run unless `allow_tests` is set in `site_config.json`.

9.1 Defects found and fixed during testing

- Select fields defaulted to their first option, so a machine-fault defect was charged to the worker. Blank first options added; defaults now come from the Defect Code.
- Work Orders created by API carried no routing, which silently skipped flow checks. Routing now falls back to the BOM operations.
- Slab incentives counted rejected pieces. They now count accepted pieces only.
- Fetching entries overwrote the batch's chosen posting mode with the settings default. Fixed.
- QC deductions could be paid ahead of a log still pending review. They now wait for approval.
- Additional Salary posting failed per employee without a Salary Structure. A pre-flight check now lists missing assignments before submit.

10. Limitations and decisions for go-live

Item	Status / action
Labour-law caps	Defaults follow commonly cited KSA Labor Law limits. Legal counsel must confirm the values and whether piece-rate damage deductions need worker acknowledgement.
Desk pages in a browser	Server load and script syntax verified; visual check of both pages in Arabic RTL on the target monitors and tablets is still required.
Attendance enforcement	Uses HRMS Attendance and Employee Checkin; not exercised in automated tests.
Employee skills and multi-skill bonus	Logic implemented; not covered by the live scenario.
Leaderboard	Publishing individual rankings can affect morale and privacy. Enable masked names or cell-only view where appropriate.
Volume	Tested functionally, not under load. Run a pilot with a full shift of device traffic before rollout.
Mid-day batches	Batch only closed days to keep daily slabs correct.